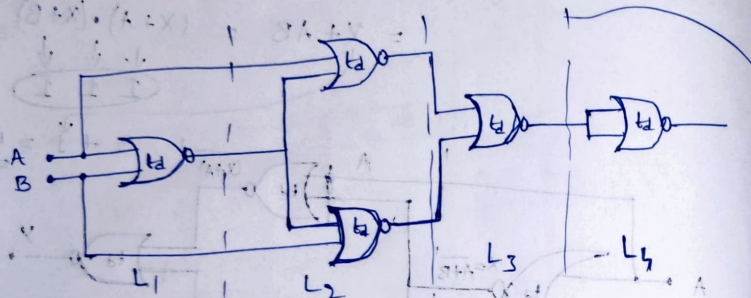


② XOR :- $y = A \oplus B$

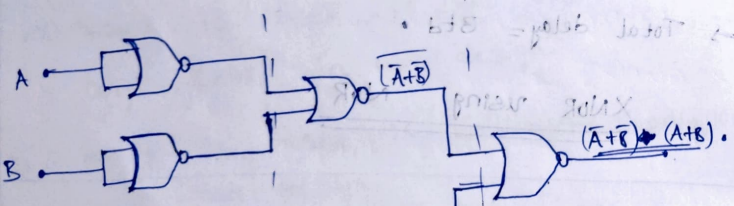
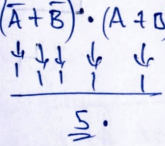
$= \overline{A}B + A\overline{B}$

⇒ Way-1 :- $A \oplus B = \overline{A \odot B}$



⇒ Total delay = $4t_d$ (4 level Assembly)

⇒ Way-2 :- $y = A \oplus B = \overline{A}B + A\overline{B} = (\overline{A+B}) \cdot (A+B)$



3-level assembly - L_1, L_2, L_3 .

END of DAY-3

DAY-4

(03-06-25)

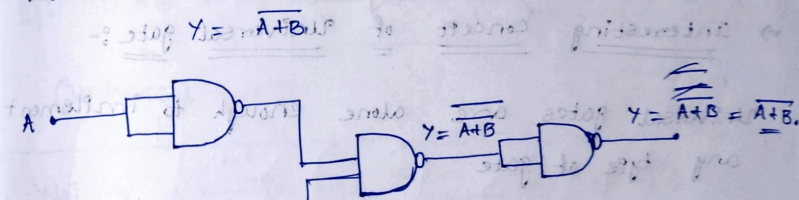
(a) 2 i/p XOR & XNOR gates are implemented using 2 i/p NOR & NAND gates, respectively. Find Min. Possible Propagation delay

A. XOR gate → NOR → 5 gates = level-3 = $3t_d$

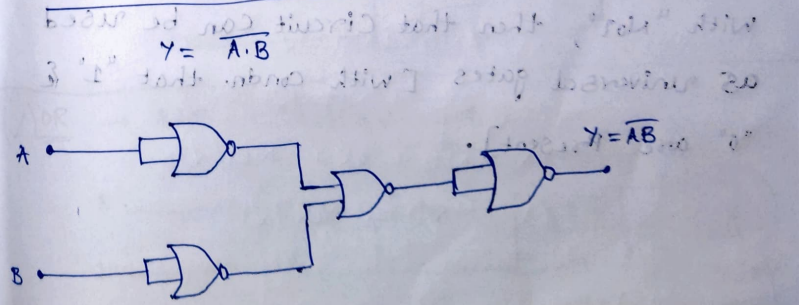
XNOR || → NAND → 5 gates = level-3 = $3t_d$

Min. delay = $3t_d$

(*) NOR gate using NAND :-



(*) NAND gate using NOR :-



(*) Min. gates required:-

	NAND	NOR
NOT	1	1
AND	2	3
OR	3	2
NOR	4	1
NAND	1	4
XOR	4	5
XNOR	5	4

(*) NAND $\rightarrow$ Alone enough to implement any Boolean Fn.

NOR $\rightarrow$ Alone enough to implement any Boolean Fn.

AND + NOT $\rightarrow$ 11

OR + NOT $\rightarrow$ 11

AND + OR + NOT $\rightarrow$ 14

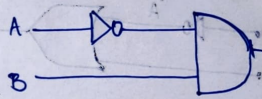
$\Rightarrow$ Interesting concept of Universal gate:-

Universal gates are alone enough to implement any type of gate

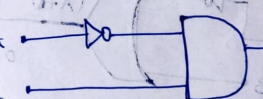
(a) Are there only 2 universal gates?

A. Generally, there are only 2 [NAND, NOR].

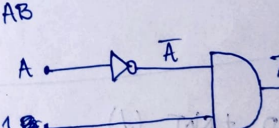
But, if any circuit can give "AND" (or) "OR" with "NOT", then that circuit can be used as universal gates [with condn. that "1" & "0" are present].

Eq:- $\bar{A}\bar{B} \Rightarrow$  $Y = \bar{A}\bar{B}$

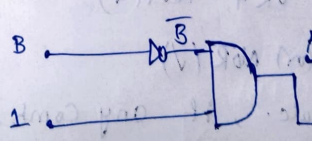
(Universal gate!)

(✓) NOT $\rightarrow$  $Y = \bar{A} \cdot 1 = \bar{A}$

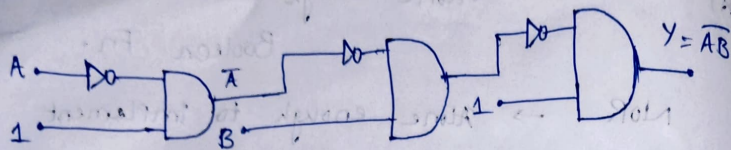
(✓) AND $\rightarrow AB$

 $Y = AB$

(✓) OR $\rightarrow A+B = \overline{\bar{A}\bar{B}} = \overline{\bar{A} \cdot \bar{B}}$

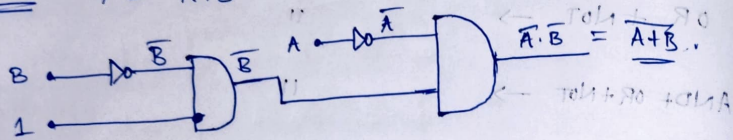
 $Y = \overline{\bar{A}\bar{B}} = A+B$

NAND:- $Y = \overline{AB}$ (✓)

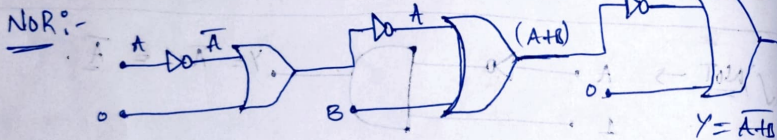


can implement any expression.

NOR:- $Y = \overline{A+B}$



② $\overline{A+B} \Rightarrow A \rightarrow \overline{A} \rightarrow \overline{A+B} = \overline{A+B}$



Conclusion:- $\Rightarrow$ AND + NOT (✓)

$\Rightarrow$ OR + NOT (✓)

$\Rightarrow$ AND + OR + NOT (✓)

$\Rightarrow$ NAND (or) NOR (✓)

$\Rightarrow$ When you do Boolean Calc., if any Comb. found in Expression, it is Universal.

(a) Which of these are universal gates?

① $\Rightarrow$ NOR (✓)

② $\Rightarrow \overline{A+B} = \overline{A+B} = \underline{\underline{A+B}}$ (X)
No Combination.

③ $\Rightarrow \overline{A+B} = \underline{\underline{A+B}}$ (✓)

④ $\Rightarrow \overline{A+B} = \underline{\underline{A+B}}$ (✓)

⑤ $\Rightarrow \overline{A+B} = \underline{\underline{A+B}}$ (✓)

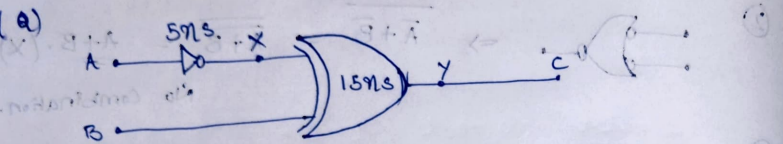
⑥ $\Rightarrow \overline{A+B} = \underline{\underline{A+B}}$ (X)
(No Combination)

(a) $Y = A+B$, what gates can be implemented using this gate?

A) NOT + OR = NOR (✓), universal gate (✓)

As given Comb. satisfies universal gate condition, any gate can be implemented like NAND, NOR, AND, OR, etc...

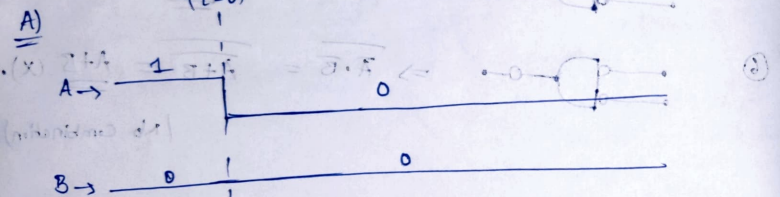
(★) Timing Diagrams of different Digital circuits involving elements which are having non-zero delay:-



For $(t < 0)$:- $A = 1, B = 0$

For $(t > 0)$:- $A = 0, B = 0$

⇒ NOT gate has 5ns delay and OR gate has 15ns delay. Draw timing diagram of C?



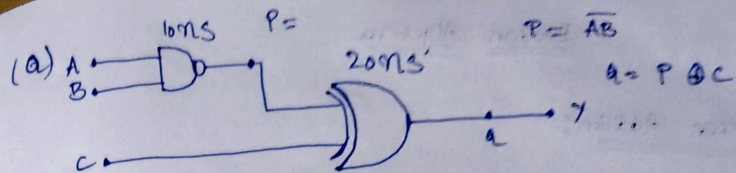
X (w.o) →

X (5ns) →

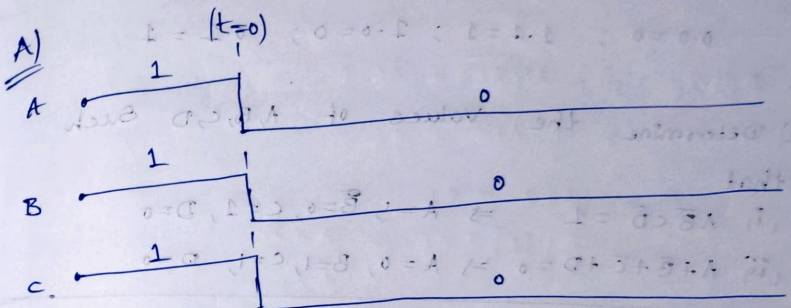
Y (w.o) →

Y (15ns) →

$$X = \bar{A} ; Y = X \oplus B$$



All i/p are changing from "1" to "0" at $(t=0)$. Draw timing diagram of Y?



(w.o)

(10ns)

(w.o)

(20ns)

Boolean Algebra

⇒ Boolean Addition:-

$$0+0=0; 1+1=1; 1+0=1; 0+1=1$$

⇒ Boolean Multiplication:-

$$0 \cdot 0 = 0; 1 \cdot 1 = 1; 1 \cdot 0 = 0; 0 \cdot 1 = 0$$

(A) Determine the values of A, B, C, D such that

(i) $A\bar{B}C\bar{D} = 1 \Rightarrow A=1, B=0, C=1, D=0$

(ii) $A+\bar{B}+\bar{C}+D=0 \Rightarrow A=0, B=1, C=1, D=0$

⇒ Laws of Boolean Algebra:-

(i) Commutative Laws:-

$$\rightarrow A+B = B+A$$

$$\rightarrow AB = BA$$

$$\rightarrow A \oplus B = B \oplus A$$

$$\rightarrow A \odot B = B \odot A$$

(ii) Associative Law:-

$$\rightarrow (A+B)+C = A+(B+C)$$

$$\rightarrow (AB)C = A(BC)$$

$$\rightarrow (A \oplus B) \oplus C = A \oplus (B \oplus C)$$

$$\rightarrow (A \odot B) \odot C = A \odot (B \odot C)$$

(iii) Distributive Law:-

$$\Rightarrow A(B+C) = AB+AC$$

$$\Rightarrow A(B \odot C) = AB \oplus AC$$

$$\Rightarrow [A \oplus (B+C)] \neq (A \oplus B) + (A \oplus C)$$

⇒ Rules of Boolean Algebra:-

$$\rightarrow A+0 = A$$

$$\rightarrow A+1 = 1$$

$$\rightarrow A+A = A$$

$$\rightarrow A+A+\dots+A = A$$

$$\rightarrow A+\bar{A} = 1$$

$$A \cdot 0 = 0$$

$$A \cdot 1 = A$$

$$A \cdot A = A$$

$$A \cdot A \cdot \dots \cdot A = A$$

$$A \cdot \bar{A} = 0$$

$$A+AB = A(1+B) = A \cdot 1 = A$$

$$(*) (A+B)(A+C) = A+AC+AB+BC$$

$$= A[1+B+C] + BC$$

$$= A+BC$$

2nd way $\Rightarrow (A+B)(A+C) = A + \underline{B \cdot C}$ (Distributive Law)

$$\Rightarrow A+BC = (A+B)(A+C)$$

$$\Rightarrow A + \bar{A}C = (A + \bar{A}) \cdot (A + C)$$

$$= (1) \cdot (A + C) = A + C$$

~~$$\Rightarrow A \oplus \bar{A}C = (A \oplus \bar{A}) \cdot (A \oplus C)$$

$$= [\bar{A} + A] \cdot (\bar{A}C + AC)$$

$$= \bar{A}C + AC$$~~

$$\Rightarrow A \oplus \bar{A}C = A(\bar{A}C) + (\bar{A}C)\bar{A}$$

$$= A(\bar{A}C) + \bar{A}C$$

$$A = A + \bar{A}C + \bar{A}C$$

$$A = A[1 + \bar{C}] + \bar{A}C = A + \bar{A}C$$

$$= \underline{A + C}$$

$$(\star) X + \bar{X}Y = X + Y$$

END OF
DAY - 4

$$A + B + C + D = (A + B)(A + C)(A + D)$$

$$A + B + C + D = (A + B + C + D)A$$

$$= A + B + C + D$$

$$(A + B + C + D)A = (A + B + C + D)(A + B + C + D)A$$

$$(A + B + C + D)A = A + B + C + D$$